ModelIt!

Student Activity Pack

Why Do You Weigh Less on the Moon?

Gravity, Mass, and the Force That Keeps You Grounded

Grade 07L2 | NGSS MS-PS2-4, MS-ESS1-2 | Systems and System Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think the difference is between mass and weight?

2 Why do you think astronauts can jump so high on the Moon?

3 Draw what you think gravity looks like pulling on you on Earth vs. on the Moon.

4 Do you think you would weigh the same on every planet? Why or why not?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Planet Mass | Distance from Center | Gravitational Pull | Object Weight | Jump Height

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Moon vs. Earth

Low planet mass (Moon), same distance from center

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Giant Planet

Very high planet mass (Jupiter), same distance from center

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Far from Center

Same planet mass, greatly increased distance from center

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Atmospheric Density — The thickness of the atmosphere surrounding the planet, which affects air resistance and terminal velocity of falling objects

Orbital Velocity — The speed needed for an object to stay in orbit around the planet, which depends on gravitational pull

Tidal Force — The difference in gravitational pull across an object's width, caused by being close to a massive body like a planet or moon

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why would you weigh less on the Moon than on Earth, even though your body hasn't changed at all? Use evidence from your simulation to support your answers.

1. **Why is it important to distinguish between mass and weight in science? How does your model make this distinction clear?**

2. **If you were designing a bathroom scale for the Moon, how would it need to be different from an Earth scale?**

3. **How does the inverse square relationship between distance and gravity affect spacecraft as they travel between planets?**

4. **Why doesn't the International Space Station fall to Earth even though it's within Earth's gravitational pull?**

5. **How does your model help explain why planets closer to the Sun orbit faster than distant planets?**

8 STEM Challenge: Design a Space Colony Fitness Program

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YOUR ENGINEERING MISSION

Create a fitness and safety program for astronauts living on the Moon or Mars, accounting for how reduced gravity affects the human body, using data from your simulation model.

THE SCENARIO

NASA is planning a permanent Moon base and needs your team to design daily fitness routines that account for the Moon's reduced gravity. Astronauts lose bone density and muscle mass in low gravity, so your program must keep them healthy for a two-year mission.

Engineering Design Thinking

1. WHAT GRAVITY LEVEL WILL YOUR COLONY EXPERIENCE, AND HOW DOES THAT COMPARE TO EARTH?
2. HOW WILL YOU SIMULATE EARTH-LIKE RESISTANCE FOR MUSCLES AND BONES IN LOW GRAVITY?
3. WHAT MODEL DATA WILL YOU USE TO CALCULATE THE DIFFERENCE IN PHYSICAL DEMANDS?
4. HOW WILL YOU MONITOR ASTRONAUT HEALTH AND ADJUST THE PROGRAM OVER TIME?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Astrophysicists study the physics of celestial objects and the forces that govern their behavior. They work at NASA, universities, observatories, and aerospace companies, earning $80,000-$160,000/year.